Swiggy Data Science Project

In [1]:

```
#Import required libraries:
import pandas as pd
import numpy as np
import joblib
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split


from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder , LabelEncoder



from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import ExtraTreesRegressor, AdaBoostRegressor
from xgboost import XGBRegressor

from sklearn.model_selection import train_test_split , cross_validate , KFold

from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import ExtraTreesClassifier , AdaBoostClassifier
from xgboost import XGBClassifier




from sklearn.tree import DecisionTreeClassifier
from sklearn.tree import plot_tree



from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix,
    r2_score,
    mean_squared_error,
    mean_absolute_error
)

from sklearn.model_selection import GridSearchCV


import warnings
warnings.filterwarnings("ignore")
```

In [ ]:

```

```

In [2]:

```
#Load the dataset:
df = pd.read_csv("dataset/swiggy.csv")
df.head()
```

Out[2]:

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
0	211	Koramangala	Bangalore	Tandoor Hut	300.0	4.4	100	Biryani,Chinese,North Indian,South Indian	5Th Block	59
1	221	Koramangala	Bangalore	Tunday Kababi	300.0	4.1	100	Mughlai,Lucknowi	5Th Block	56
2	246	Jogupalya	Bangalore	Kim Lee	650.0	4.4	100	Chinese	Double Road	50
3	248	Indiranagar	Bangalore	New Punjabi Hotel	250.0	3.9	500	North Indian,Punjabi,Tandoor,Chinese	80 Feet Road	57
4	249	Indiranagar	Bangalore	Nh8	350.0	4.0	50	Rajasthani,Gujarati,North Indian,Snacks,Desser...	80 Feet Road	63

In [3]:

```
#Check for any null-values:
df.isnull().sum()
```

Out[3]:

```
ID                0
Area              0
City              0
Restaurant        0
Price             0
Avg ratings       0
Total ratings     0
Food type         0
Address           0
Delivery time     0
dtype: int64
```

In [4]:

```
#Check for any duplicate values:
df.duplicated().sum()
```

Out[4]:

```
np.int64(0)
```

In [5]:

```
df.describe()
```

Out[5]:

	ID	Price	Avg ratings	Total ratings	Delivery time
count	8680.000000	8680.000000	8680.000000	8680.000000	8680.000000
mean	244812.071429	348.444470	3.655104	156.634793	53.967051
std	158671.617188	230.940074	0.647629	391.448014	14.292335
min	211.000000	0.000000	2.000000	20.000000	20.000000
25%	72664.000000	200.000000	2.900000	50.000000	44.000000
50%	283442.000000	300.000000	3.900000	80.000000	53.000000
75%	393425.250000	400.000000	4.200000	100.000000	64.000000
max	466928.000000	2500.000000	5.000000	10000.000000	109.000000

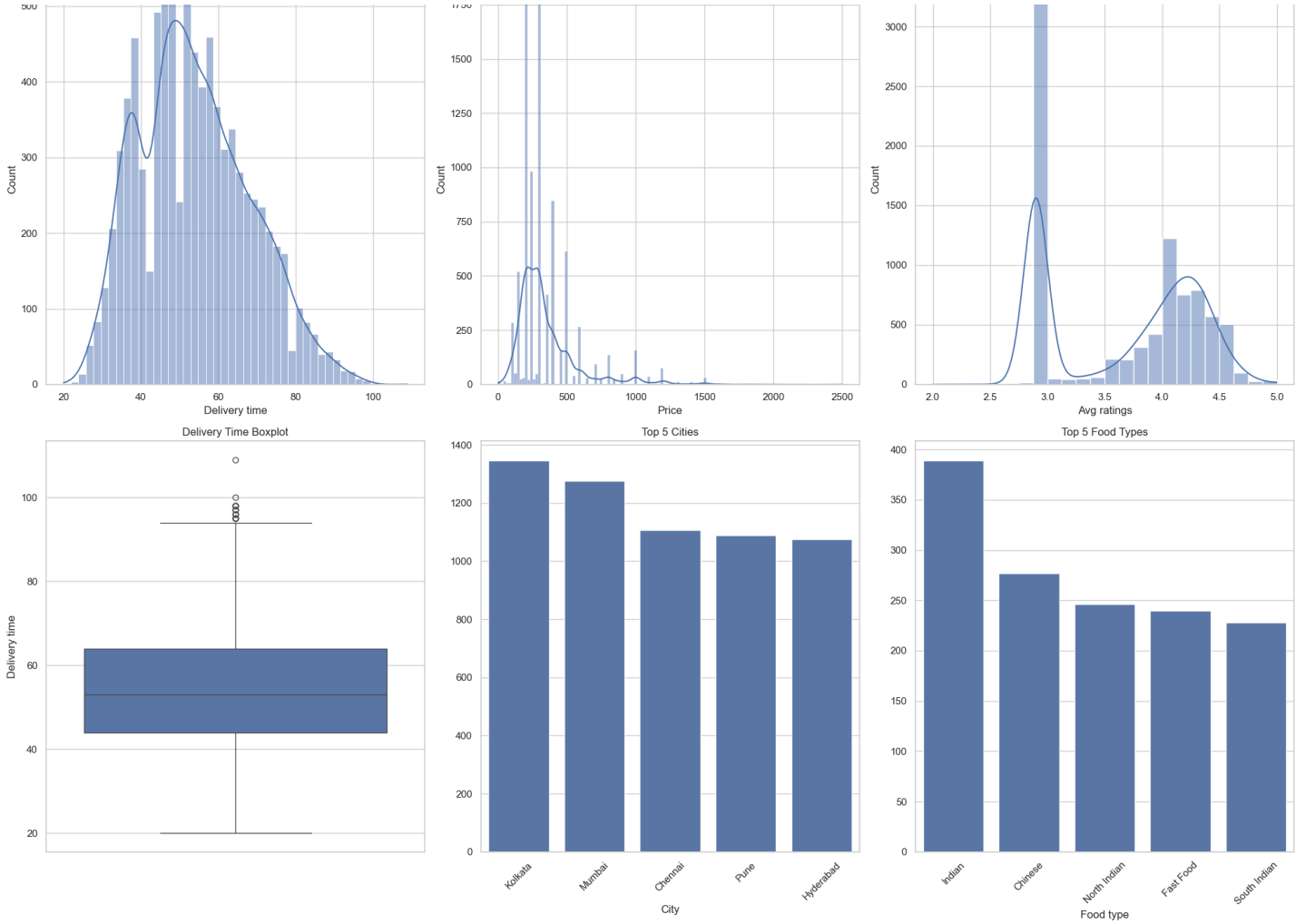
In [6]:

```
#View basic information about the dataset:
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8680 entries, 0 to 8679
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   ID             8680 non-null   int64  
 1   Area           8680 non-null   object 
 2   City           8680 non-null   object 
 3   Restaurant     8680 non-null   object 
 4   Price          8680 non-null   float64
 5   Avg ratings    8680 non-null   float64
 6   Total ratings  8680 non-null   int64  
 7   Food type      8680 non-null   object 
 8   Address        8680 non-null   object 
 9   Delivery time  8680 non-null   int64  
dtypes: float64(2), int64(3), object(5)
memory usage: 678.3+ KB
```

In [7]:

```
# Basic EDA:


# Set style
sns.set(style="whitegrid")

# Create 2x3 subplot grid
fig, axes = plt.subplots(2, 3, figsize=(20, 15))

# ========== ROW 1 ==========
# 1. Delivery Time Distribution
sns.histplot(df['Delivery time'], kde=True, ax=axes[0, 0])
axes[0, 0].set_title("Delivery Time Distribution")

# 2. Price Distribution
sns.histplot(df['Price'], kde=True, ax=axes[0, 1])
axes[0, 1].set_title("Price Distribution")

# 3. Avg Ratings Distribution
sns.histplot(df['Avg ratings'], kde=True, ax=axes[0, 2])
axes[0, 2].set_title("Average Ratings Distribution")



# ========== ROW 2 ==========
# 4. Boxplot of Delivery Time
sns.boxplot(y=df['Delivery time'], ax=axes[1, 0])
axes[1, 0].set_title("Delivery Time Boxplot")

# 5. Top 5 Cities Count
top_cities = df['City'].value_counts().head(5)
sns.barplot(x=top_cities.index, y=top_cities.values, ax=axes[1, 1])
axes[1, 1].set_title("Top 5 Cities")
axes[1, 1].tick_params(axis='x', rotation=45)

# 6. Top 5 Food Types Count
top_food_types = df['Food type'].value_counts().head(5)
sns.barplot(x=top_food_types.index, y=top_food_types.values, ax=axes[1, 2])
axes[1, 2].set_title("Top 5 Food Types")
axes[1, 2].tick_params(axis='x', rotation=45)


# Adjust layout
plt.tight_layout()
plt.show()
```



Hypothesis Testing Questions

Q1] Swiggy wants to determine whether the average delivery time of restaurants on its platform differs significantly from the company's target delivery time of 45 minutes. (one sample ztest)

In [8]:

```
from statsmodels.stats.weightstats import ztest
z_stat , p_value = ztest(df["Delivery time"],value=45)

if p_value < 0.05:
    print("Reject Null Hypothesis: Average delivery time is significantly different from 45 minutes.")
else:
    print("Fail to Reject Null Hypothesis: Average delivery time is not different from 45 minutes.")
```

```
Reject Null Hypothesis: Average delivery time is significantly different from 45 minutes.
```

Q2] Swiggy wants to determine whether Indian restaurants and Chinese restaurants have significantly different average delivery times. (two sample ztest)

In [9]:

```
indian_delivery = df[df['Food type'] == 'Indian']['Delivery time']
chinese_delivery = df[df['Food type'] == 'Chinese']['Delivery time']

z_stat, p_value = ztest(
    indian_delivery,
    chinese_delivery,
    alternative='two-sided'
)

if p_value < 0.05:
```

```
    print("Reject the Null Hypothesis: Indian and Chinese restaurants have significantly different average delivery times.")
else:
    print("Fail to Reject the Null Hypothesis: There is no significant difference in the average delivery times of Indian and Chinese restaurants.")
```

```
Fail to Reject the Null Hypothesis: There is no significant difference in the average del
ivery times of Indian and Chinese restaurants.
```

Q3] Swiggy wants to determine whether the average customer rating of restaurants on its platform is significantly different from 4.0, which is considered the company's benchmark for a well-performing restaurant. (one sample ttest)

In [10]:

```
#Since the population standard deviation of restaurant ratings is unknown, t-Test is used.

from scipy.stats import ttest_1samp

t_stat , p_value = ttest_1samp(a=df["Avg ratings"], popmean=4)

if p_value < 0.05:
    print("Average customer rating is significantly different from 4")
else:
    print("Average customer rating is 4")
```

```
Average customer rating is significantly different from 4
```

Q4] Swiggy wants to determine whether Indian restaurants and Chinese restaurants have significantly different average customer ratings. (two sample ttest)

In [11]:

```
from scipy.stats import ttest_ind

indian_ratings = df[df['Food type'] == 'Indian']['Avg ratings']
chinese_ratings = df[df['Food type'] == 'Chinese']['Avg ratings']

t_stat , p_value = ttest_ind(a=indian_ratings, b=chinese_ratings, equal_var=False)

if p_value < 0.05:
    print("Both restaurants have significantly different average customer ratings.")
else:
    print("Both restaurants have same average customer ratings.")
```

```
Both restaurants have significantly different average customer ratings.
```

Q5] Swiggy wants to determine whether the average delivery time differs significantly among Indian, Chinese, and Fast Food restaurants. (one-way anova)

In [12]:

```
from scipy.stats import f_oneway

#Boolean masks of all food types:
indian_mask = df['Food type'].str.contains('Indian', case=False, na=False)
chinese_mask = df['Food type'].str.contains('Chinese', case=False, na=False)
fast_food_mask = df['Food type'].str.contains('Fast Food', case=False, na=False)

#Delivery time obtained from food-types:
indian = df.loc[indian_mask, 'Delivery time']
chinese = df.loc[chinese_mask, 'Delivery time']
fast_food = df.loc[fast_food_mask, 'Delivery time']

f_stat, p_value = f_oneway(indian, chinese, fast_food)


if p_value < 0.05:
```

```
    print("Reject the Null Hypothesis: At least one cuisine type has a significantly diff
erent average delivery time.")
else:
    print("Fail to Reject the Null Hypothesis: There is no significant difference in aver
age delivery time among the cuisine types.")
```

```
Reject the Null Hypothesis: At least one cuisine type has a significantly different avera
ge delivery time.
```

Q6] Swiggy wants to determine whether restaurant cuisine type is associated with the city in which the restaurant operates. (Chi-Square Test of Independence)

In [13]:

```
from scipy.stats import chi2_contingency


table = pd.crosstab(df['City'], df['Food type'])

chi2, p_value, dof, expected = chi2_contingency(table)


if p_value < 0.05:
    print("Reject the Null Hypothesis: Food type is significantly associated with the cit
y.")
else:
    print("Fail to Reject the Null Hypothesis: Food type and city are independent.")
```

```
Reject the Null Hypothesis: Food type is significantly associated with the city.
```

Machine Learning Questions

Q1] Predict the Delivery time of an order using restaurant, pricing, ratings, and location features.

In [14]:

```
#Feature / Target Split:
X = df.drop(columns=['Delivery time', 'ID', 'Food type'])
y = df['Delivery time']



X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

#Preprocessing:
num_cols = X_train.select_dtypes(include=['int64', 'float64']).columns
cat_cols = X_train.select_dtypes(include=['object']).columns

preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_cols),
    ('cat', OneHotEncoder(handle_unknown='ignore'), cat_cols)
])
```

In [15]:

```
#Linear Regression:
lr_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', LinearRegression())
])


#KNN Regressor:
knn_pipeline = Pipeline([
    ('preprocessing', preprocessor),
```

```
    ('model', KNeighborsRegressor())
])


#Decision-tree Regressor:
dt_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', DecisionTreeRegressor(random_state=42))
])


#Extra-trees Regressor:
et_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', ExtraTreesRegressor(n_estimators=250, max_depth=5, random_state=42))
])


#Adaboost:
ada_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', AdaBoostRegressor(
        estimator=DecisionTreeRegressor(max_depth=7),
        n_estimators=400,
        learning_rate=0.5,
        random_state=42
    ))
])


#XG-Boost:
xgb_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', XGBRegressor(
        n_estimators=500,
        learning_rate=0.05,
        max_depth=6,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
        n_jobs=-1
    ))
])
```

In [16]:

```
#Model dictionary:
models = {
    "Linear Regression": lr_pipeline,
    "KNN": knn_pipeline,
    "Decision Tree": dt_pipeline,
    "Extra Trees": et_pipeline,
    "AdaBoost": ada_pipeline,
    "XGBoost":xgb_pipeline
}
```

In [17]:

```
#Cross validation:


kf = KFold(n_splits=5, shuffle=True, random_state=42)

scoring = {
    'r2': 'r2',
    'mae': 'neg_mean_absolute_error',
    'rmse': 'neg_root_mean_squared_error'
}

results = []
```

```
for name, model in models.items():

    cv_results = cross_validate(
        model,
        X_train,
        y_train,
        cv=kf,
        scoring=scoring
    )

    mae = -np.mean(cv_results['test_mae'])
    rmse = -np.mean(cv_results['test_rmse'])
    r2 = np.mean(cv_results['test_r2'])

    results.append({
        "Model": name,
        "CV R2": round(r2, 4),
        "CV MAE": round(mae, 2),
        "CV RMSE": round(rmse, 2)
    })

results_df = pd.DataFrame(results).sort_values(by="CV R2", ascending=False)
print(results_df)
```

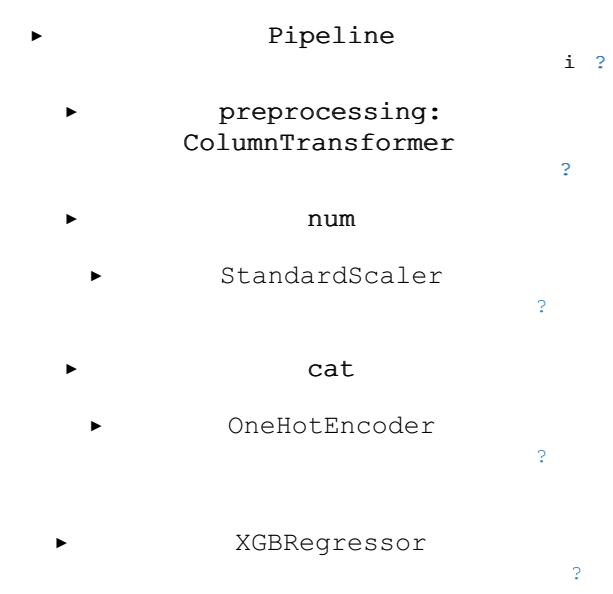
```
               Model   CV R2  CV MAE  CV RMSE
5            XGBoost  0.6644    6.33     8.25
1                KNN  0.6505    6.27     8.42
2      Decision Tree  0.5216    7.22     9.85
0  Linear Regression  0.4763    7.69    10.26
3        Extra Trees  0.3527    9.18    11.46
4           AdaBoost  0.3438    9.47    11.54
```

In [18]:

```
#Train the best model on Full-training data:
best_model = results_df.iloc[0]['Model']
print("Best Model:", best_model)

best_pipeline = models[best_model]
best_pipeline.fit(X_train, y_train)
```

```
Best Model: XGBoost
```

Out[18]:



In [19]:

```
#Test Evaluation on testing:


y_pred = best_pipeline.predict(X_test)
```

```
print("TEST RESULTS")
print("R2:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
TEST RESULTS
R2: 0.675024151802063
MAE: 6.412059307098389
RMSE: 8.241580250142036
```

In [20]:

```
#Perform Hyperparamter tuning on the best pipeline:


param_grid = {
    'model__n_estimators': [100, 200, 300],
    'model__max_depth': [3, 5, 7],
    'model__learning_rate': [0.01, 0.05, 0.1],
    'model__subsample': [0.8, 1.0],
    'model__colsample_bytree': [0.8, 1.0]
}

grid_search = GridSearchCV(
    estimator=best_pipeline,
    param_grid=param_grid,
    scoring='neg_root_mean_squared_error',
    cv=5,
    n_jobs=-1,
    verbose=2
)

grid_search.fit(X_train, y_train)


rmse = mean_squared_error(y_test, y_pred) ** 0.5
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("RMSE:",rmse)
print("MAE :",mae)
print("R²  :",r2)
```

```
Fitting 5 folds for each of 108 candidates, totalling 540 fits
RMSE: 8.241580250142036
MAE : 6.412059307098389
R²  : 0.675024151802063
```

Even after doing hyper-parameter tuning no change was observed in the values of RMSE, R2 Score and MAE. Hence we will not use hyper-parameter tuning in this case.

In [21]:

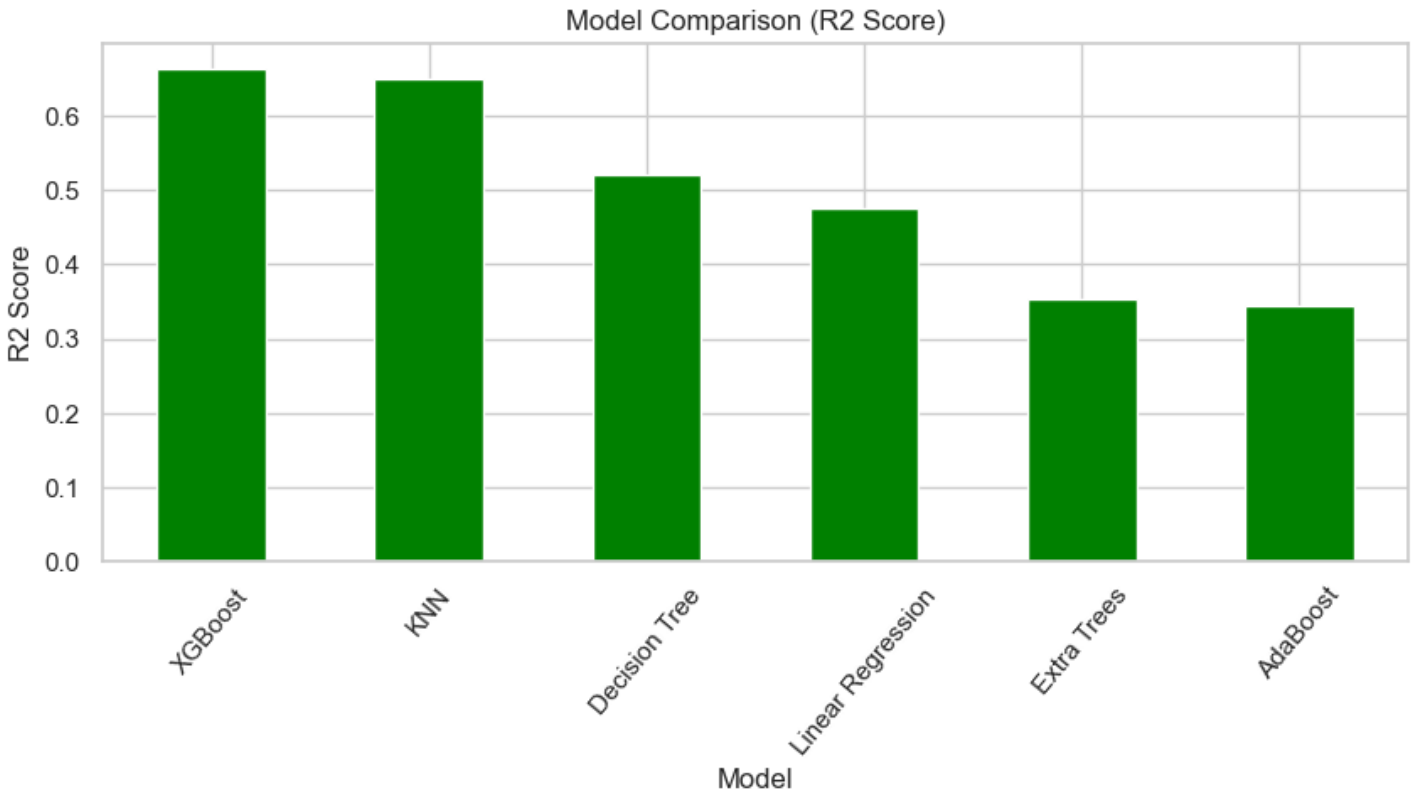
```
#Now test on one single tuple:
single_data = pd.DataFrame({
    'Area': ['Vastrapur'],
    'City': ['Ahmedabad'],
    'Restaurant': ['Vanilla Sky'],
    'Price': [300.0],
    'Avg ratings': [2.9],
    'Address': ['Vastrapur'],
    'Total ratings': [220]
})

pred = best_pipeline.predict(single_data)
print("Predicted Delivery Time:", round(pred[0]), "minutes")
```

```
Predicted Delivery Time: 48 minutes
```

In [ ]:

In [22]:

```
#Model comparison:

plt.figure(figsize=(10,4))
results_df.set_index("Model")["CV R2"].plot(kind='bar',color="green")
plt.title("Model Comparison (R2 Score)")

plt.xlabel("Model")
plt.ylabel("R2 Score")
plt.xticks(rotation=50)
plt.show()
```



In [ ]:

In [23]:

```
#Export the best-performing model:

joblib.dump(grid_search,"Swiggy_FoodDeliveryTimePrediction_Model.pkl")
```

Out[23]:

```
['Swiggy_FoodDeliveryTimePrediction_Model.pkl']
```

In [ ]:

CONCLUSION : We used several traditional-ml algorithms to predict the delivery time. The predicted delivery time is 48 minutes.

In [ ]:

In [ ]:

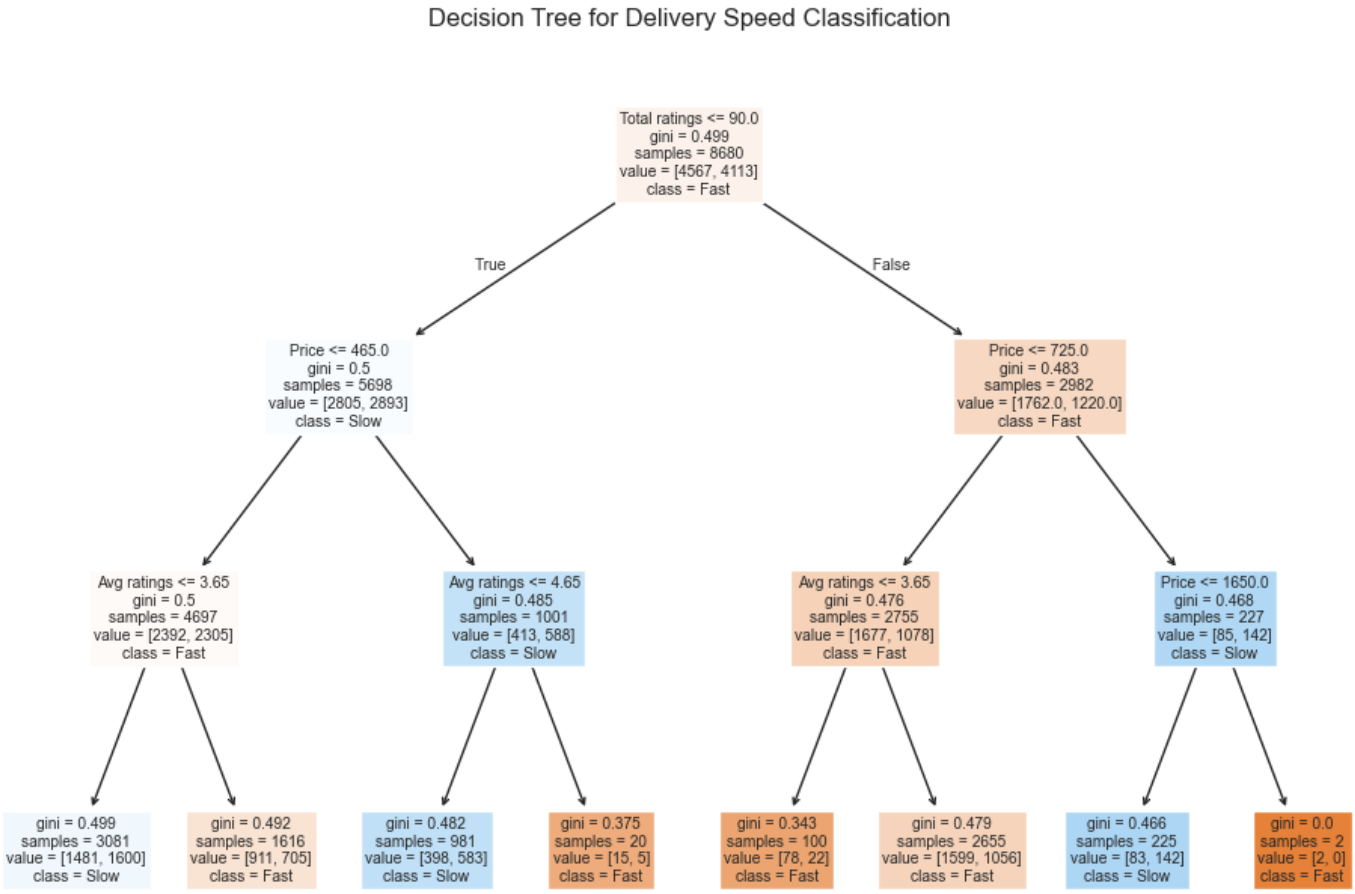
In [ ]:

In [ ]:

Q2] Can we classify whether a restaurant will have fast or slow delivery based on its price, ratings, and popularity?

In [24]:

```
df['delivery_speed'] = df['Delivery time'].apply(
    lambda x: 'Fast' if x <= df['Delivery time'].median() else 'Slow'
)


X = df[['Price', 'Avg ratings', 'Total ratings']]
y = df['delivery_speed']




model = DecisionTreeClassifier(max_depth=3, random_state=42)
model.fit(X, y)



plt.figure(figsize=(12,8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Fast', 'Slow'],
    filled=True
)
plt.title("Decision Tree for Delivery Speed Classification")
plt.show()
```



CONCLUSION]

- **Expensive Restaurants tend to deliver food faster.**
- **Restaurants having a more preference (based on average user ratings) tend to deliver food faster.**
- **Restaurants having low ratings (based on average user ratings) tend to deliver food slower.**

In [25]:

```
df.head()
```

Out[25]:

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time	delive
0	211	Koramangala	Bangalore	Tandoor Hut	300.0	4.4	100	Biryani,Chinese,North Indian,South Indian	5Th Block	59	
1	221	Koramangala	Bangalore	Tunday Kababi	300.0	4.1	100	Mughlai,Lucknowi	5Th Block	56	
2	246	Jogupalya	Bangalore	Kim Lee	650.0	4.4	100	Chinese	Double Road	50	
3	248	Indiranagar	Bangalore	New Punjabi Hotel	250.0	3.9	500	North Indian,Punjabi,Tandoor,Chinese	80 Feet Road	57	
4	249	Indiranagar	Bangalore	Nh8	350.0	4.0	50	Rajasthani,Gujarati,North Indian,Snacks,Desser...	80 Feet Road	63	

In [ ]:

```
```

Q3] Predict whether a restaurant belongs to Premium or Budget segment using features like Avg Ratings, Total Ratings, Food Type, and Delivery Time (Binary Classification).

In [26]:

```
df['price_segment'] = df['Price'].apply(lambda x: 'Premium' if x > 500 else 'Budget')

print("Check for imbalance in 'price_segment' attribute:")
print(df.price_segment.value_counts())
print("\n\n\n\n\n\n")



X = df.drop(columns=['price_segment','Price', 'ID', 'Delivery time', 'delivery_speed'])
y = df['price_segment']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)



num_cols = X_train.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X_train.select_dtypes(include=['object']).columns.tolist()

preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_cols),
    ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=True), cat_cols)
])


le = LabelEncoder()
```

```
y_train_encoded = le.fit_transform(y_train)
y_test_encoded = le.transform(y_test)

# All CLassification Pipelines:
lr_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', LogisticRegression(max_iter=1000, random_state=42))
])

knn_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', KNeighborsClassifier(n_neighbors=7, n_jobs=-1))
])



et_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', ExtraTreesClassifier(n_estimators=120, max_depth=9, n_jobs=-1,random_state
=42))
])


ada_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', AdaBoostClassifier(
        estimator=DecisionTreeClassifier(max_depth=4),
        n_estimators=200,
        learning_rate=0.8,
        random_state=42
    ))
])

xgb_pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('model', XGBClassifier(
        n_estimators=300,
        learning_rate=0.1,
        max_depth=6,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
        n_jobs=-1,
        tree_method='hist',
        eval_metric='logloss'
    ))
])




# Cross Validation:
## Model Dictionary:

models = {
    "Logistic Regression": lr_pipeline,
    "KNN Classifier": knn_pipeline,
    "Extra Trees": et_pipeline,
    "AdaBoost": ada_pipeline,
    "XGBoost": xgb_pipeline
}

kf = KFold(n_splits=3, shuffle=True, random_state=42)    # 3 folds = faster
scoring = ['accuracy', 'f1_weighted']

results = []

for name, model in models.items():
    print(f"Running {name}...")

    cv_results = cross_validate(
```

```
        model,
        X_train,
        y_train_encoded,
        cv=kf,
        scoring=scoring,
        n_jobs=-1
    )

    results.append({
        "Model": name,
        "CV Accuracy": round(np.mean(cv_results['test_accuracy']), 4),
        "CV F1 Score": round(np.mean(cv_results['test_f1_weighted']), 4)
    })

results_df = pd.DataFrame(results).sort_values(by="CV Accuracy", ascending=False)
print("\n" + "="*50)
print(results_df)
```

```
Check for imbalance in 'price_segment' attribute:
price_segment
Budget     7665
Premium    1015
Name: count, dtype: int64
```

```
Running Logistic Regression...
Running KNN Classifier...
Running Extra Trees...
Running AdaBoost...
Running XGBoost...

==================================================
                Model  CV Accuracy  CV F1 Score
0  Logistic Regression       0.8895       0.8480
3             AdaBoost       0.8871       0.8405
2          Extra Trees       0.8831       0.8282
4              XGBoost       0.8823       0.8472
1       KNN Classifier       0.8774       0.8434
```

In [27]:

```
#Perform Hyperparamter tuning on the best pipeline:


param_grid = {
    'model__C': [0.001, 0.01, 0.1, 1, 10, 100],
    'model__penalty': ['l1', 'l2'],
    'model__solver': ['liblinear']
}

grid_search = GridSearchCV(
    estimator=lr_pipeline,
    param_grid=param_grid,
    scoring='accuracy',
    cv=5,
    n_jobs=-1,
    verbose=1
)

grid_search.fit(X_train, y_train)



best_model = grid_search.best_estimator_

y_pred = best_model.predict(X_test)
```

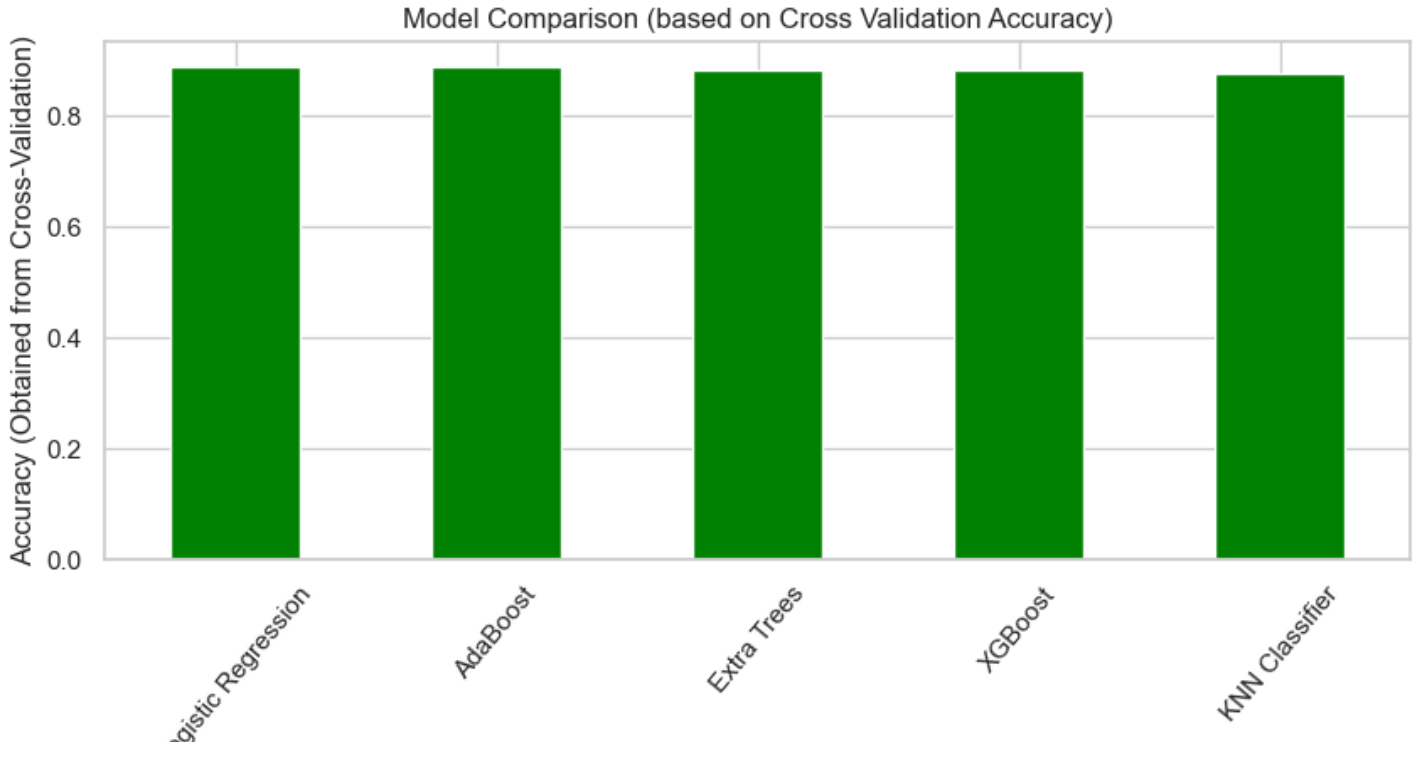
```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, pos_label='Premium'))
print("Recall:", recall_score(y_test, y_pred, pos_label='Premium'))
print("F1 Score:", f1_score(y_test, y_pred, pos_label='Premium'))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Fitting 5 folds for each of 12 candidates, totalling 60 fits
Accuracy: 0.8894009216589862
Precision: 0.6122448979591837
Recall: 0.1477832512315271
F1 Score: 0.23809523809523808

Confusion Matrix:
[[1514   19]
 [ 173   30]]

Classification Report:
              precision    recall  f1-score   support

      Budget       0.90      0.99      0.94      1533
     Premium       0.61      0.15      0.24       203

    accuracy                           0.89      1736
   macro avg       0.75      0.57      0.59      1736
weighted avg       0.86      0.89      0.86      1736
```

For the best pipeline, after doing hyper-parameter tuning accuracy decreased. Hence we will reject the hyper-paramter tuning.

In [28]:

```
#Model comparison:
plt.figure(figsize=(10,4))
results_df.set_index("Model")["CV Accuracy"].plot(kind='bar',color="green")
plt.title("Model Comparison (based on Cross Validation Accuracy)")

plt.xlabel("Model")
plt.ylabel("Accuracy (Obtained from Cross-Validation)")
plt.xticks(rotation=50)
plt.show()
```



Model

In [29]:

```
#Now test on one single tuple:
# ================= Test on One Single Tuple =================
single_data = pd.DataFrame({
    'Area': ['Ashok Nagar'],
    'City': ['Bangalore'],
    'Restaurant': ['Roomali'],
    'Avg ratings': [4.4],
    'Total ratings': [100],
    'Food type': ['North Indian,Indian'],
    'Address': ['Church Street']
})

# Make prediction using the best model
lr_pipeline.fit(X_train, y_train)
pred = lr_pipeline.predict(single_data)
print("Predicted Price Segment:", pred[0])
print("Best Model Name : ",grid_search)
```

```
Predicted Price Segment: Budget
Best Model Name :  GridSearchCV(cv=5,
             estimator=Pipeline(steps=[('preprocessing',
                                        ColumnTransformer(transformers=[('num',
                                                                         StandardScaler
(),
                                                                         ['Avg '
                                                                          'ratings',
                                                                          'Total '
                                                                          'ratings']),
                                                                        ('cat',
                                                                         OneHotEncoder(
handle_unknown='ignore'),
                                                                         ['Area',
                                                                          'City',
                                                                          'Restaurant',
                                                                          'Food '
                                                                          'type',
                                                                          'Address'])])
),
                                       ('model',
                                        LogisticRegression(max_iter=1000,
                                                           random_state=42))]),
             n_jobs=-1,
             param_grid={'model__C': [0.001, 0.01, 0.1, 1, 10, 100],
                         'model__penalty': ['l1', 'l2'],
                         'model__solver': ['liblinear']},
             scoring='accuracy', verbose=1)
```

In [30]:

```
#Observe the specifications in the 2nd record:
df.iloc[50:]
```

Out[30]:

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
50	2839	Ashok Nagar	Bangalore	C Sharp Kitchen	300.0	2.9	80	Chinese,Continental,Fast Food	Church Street	33
51	2840	Ashok Nagar	Bangalore	Roomali	1000.0	4.4	100	North Indian,Indian	Church Street	35
52	2843	Ashok Nagar	Bangalore	Tadka Singh	300.0	4.2	100	Punjabi,North Indian	Church Street	38
53	2914	Ashok Nagar	Bangalore	Bheemas	550.0	4.3	100	Andhra	Church Street	31

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
54	2915	Vasanth Nagar	Bangalore	Watson'S	450.0	2.9	80	Fast Food	Old Mass Road	58
...	...	...	...	...	...	...	...	...	...	...
8675	464626	Panjarapole Cross Road	Ahmedabad	Malt Pizza	500.0	2.9	80	Pizzas	Navrangpura	40
8676	465835	Rohini	Delhi	Jay Mata Ji Home Kitchen	200.0	2.9	80	South Indian	Rohini	28
8677	465872	Rohini	Delhi	Chinese Kitchen King	150.0	2.9	80	Chinese,Snacks,Tandoor	Rohini	58
8678	465990	Rohini	Delhi	Shree Ram Paratha Wala	150.0	2.9	80	North Indian,Indian,Snacks	Rohini	28
8679	466488	Navrangpura	Ahmedabad	Sassy Street	250.0	2.9	80	Chaat,Snacks,Chinese	Navrangpura	44

8630 rows × 12 columns

In [31]:

```
#Export the best-performing model:

joblib.dump(grid_search,"Price_Classification.pkl")
```

Out[31]:

```
['Price_Classification.pkl']
```

In [267]:

```
#Export both the models to the S3-bucket on AWS:

import boto3
import os

# Upload model to S3
best_model_name = "Price_Classification.pkl"

bucket_name = "swiggydeliverytimeandpriceclassificationmodels"
s3_folder = "models_deployed"
s3_key = s3_folder + "/"+best_model_name


#Create our S3 Client:
# Create S3 client

s3 = boto3.client(
    "s3",
    aws_access_key_id=os.getenv("AWS_ACCESS_KEY_ID"),
    aws_secret_access_key=os.getenv("AWS_SECRET_ACCESS_KEY"),
    region_name=os.getenv("AWS_REGION")
)


s3.upload_file(
    Filename=best_model_name,
    Bucket=bucket_name,
    Key=s3_key
)

print("Model uploaded successfully to S3!")
print(f"S3 Path: s3://{bucket_name}/{s3_key}")
```

```
# Upload model to S3
best_model_name = "Swiggy_FoodDeliveryTimePrediction_Model.pkl"

bucket_name = "swiggydeliverytimeandpriceclassificationmodels"
s3_folder = "models_deployed"
s3_key = s3_folder + "/"+best_model_name


#Create our S3 Client:
# Create S3 client

s3 = boto3.client(
    "s3",
    aws_access_key_id=os.getenv("AWS_ACCESS_KEY_ID"),
    aws_secret_access_key=os.getenv("AWS_SECRET_ACCESS_KEY"),
    region_name="ap-south-1"    # Mumbai region, change if needed
)



s3.upload_file(
    Filename=best_model_name,
    Bucket=bucket_name,
    Key=s3_key
)

print("Model uploaded successfully to S3!")
print(f"S3 Path: s3://{bucket_name}/{s3_key}")
```

```
Model uploaded successfully to S3!
S3 Path: s3://swiggydeliverytimeandpriceclassificationmodels/models_deployed/Price_Classi
fication.pkl
Model uploaded successfully to S3!
S3 Path: s3://swiggydeliverytimeandpriceclassificationmodels/models_deployed/Swiggy_FoodD
eliveryTimePrediction_Model.pkl
```

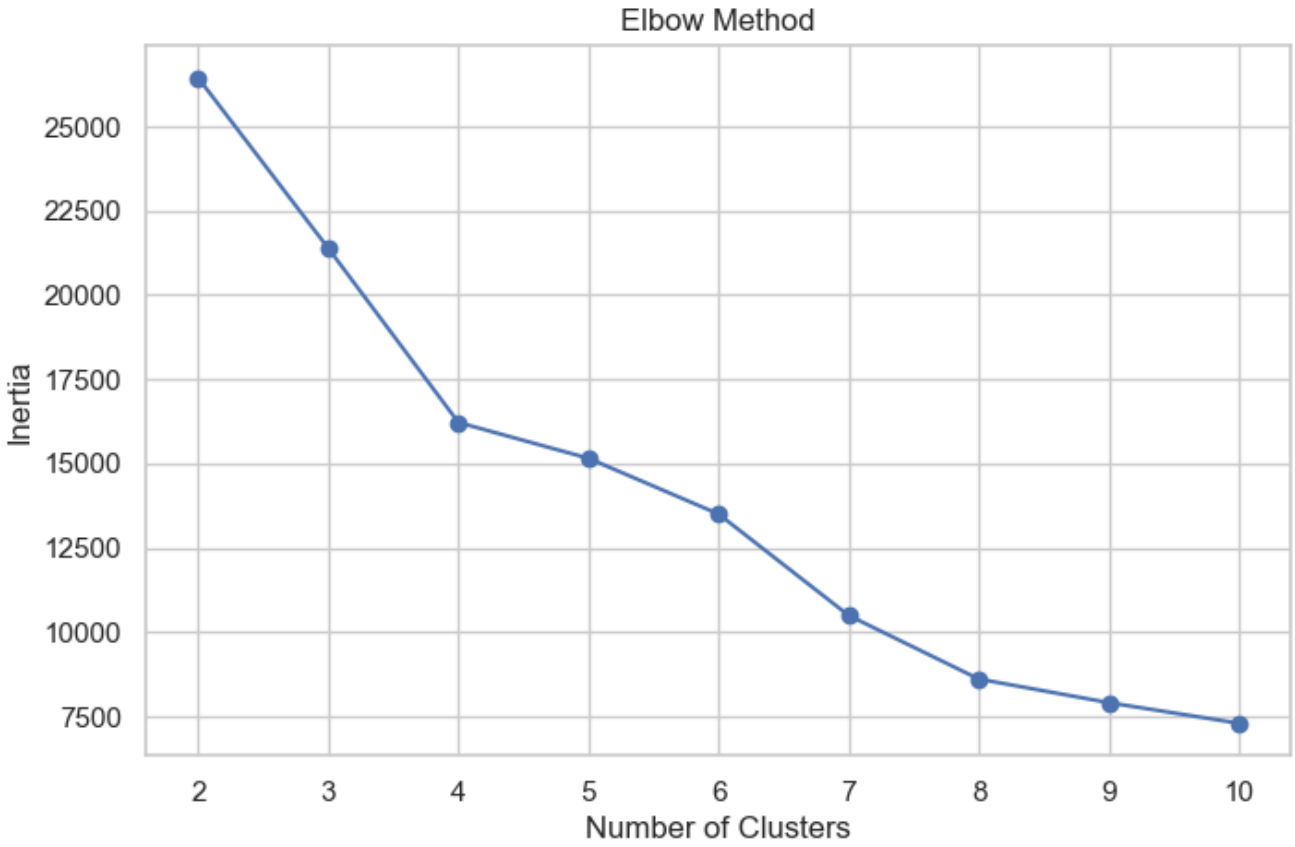
In [ ]:

```
```

Q4] Can Swiggy automatically segment restaurants into different business groups based on pricing, customer ratings, popularity, and delivery performance to support targeted marketing and business strategies?

In [34]:

```
#Find optimal no. of clusters using elbow method:
inertia = []

for k in range(2,11):
    pipeline.set_params(kmeans__n_clusters=k)
    pipeline.fit(X)
    inertia.append(
        pipeline.named_steps['kmeans'].inertia_
    )

plt.figure(figsize=(8,5))

plt.plot(range(2,11), inertia, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.grid(True)

plt.show()
```



In [35]:

```
#Find optimal no. of clusters using sihoutte score:
from sklearn.metrics import silhouette_score

silhouette_scores = []

for k in range(2, 11):

    pipeline.set_params(kmeans__n_clusters=k)

    labels = pipeline.fit_predict(X)

    score = silhouette_score(
        pipeline.named_steps['scaler'].transform(X),
        labels
    )

    silhouette_scores.append(score)

    print(f"K = {k} --> Silhouette Score = {score:.4f}")
```

```
K = 2 --> Silhouette Score = 0.3130
K = 3 --> Silhouette Score = 0.3467
K = 4 --> Silhouette Score = 0.3555
K = 5 --> Silhouette Score = 0.3493
K = 6 --> Silhouette Score = 0.3479
K = 7 --> Silhouette Score = 0.3501
K = 8 --> Silhouette Score = 0.3509
K = 9 --> Silhouette Score = 0.3469
K = 10 --> Silhouette Score = 0.3222
```

We checked for optimal no. of clusters after creating the classification pipeline only. The difference is that we wrote the code snippet on above the classification pipeline. Kindly refer this comment for any query.

In [33]:

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

X = df[['Price',
```

```
        'Avg ratings',
        'Total ratings',
        'Delivery time']]

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('kmeans', KMeans(n_clusters=4,random_state=42))
])

pipeline.fit(X)

df['Restaurant Segment'] = pipeline.predict(X)

cluster_summary = df.groupby('Restaurant Segment')[[
    'Price',
    'Avg ratings',
    'Total ratings',
    'Delivery time'
]].mean()

cluster_summary
```

Out[33]:

Restaurant Segment	Price	Avg ratings	Total ratings	Delivery time
0	302.198659	4.167531	198.556425	50.885587
1	288.701526	2.953646	75.197852	57.081967
2	995.955178	3.934158	115.919629	58.544049
3	317.500000	4.060000	6500.000000	44.350000

In [36]:

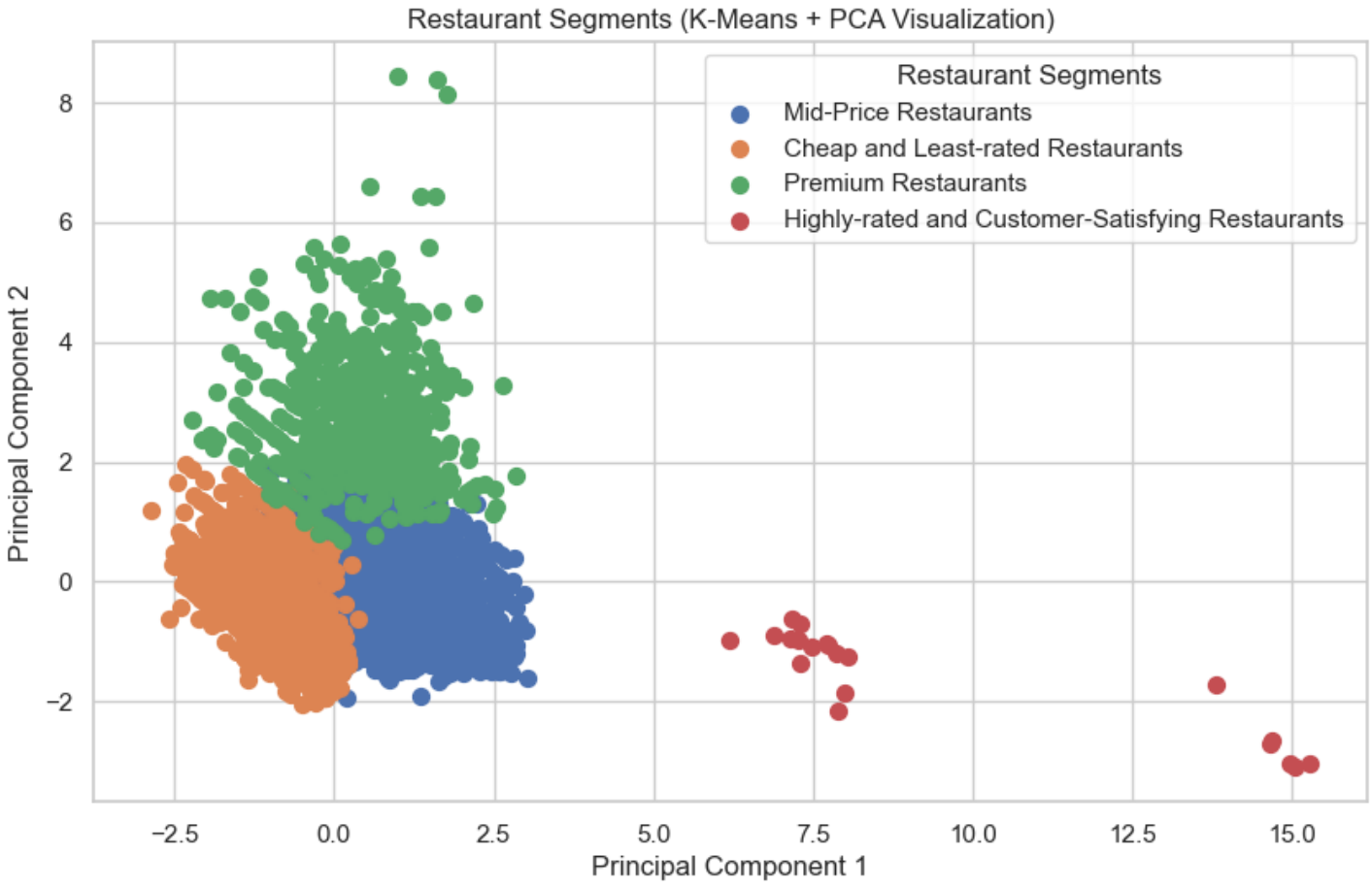
```
print("No. of restaurants in each segment = ",df['Restaurant Segment'].value_counts())
```

```
No. of restaurants in each segment =  Restaurant Segment
0    4475
1    3538
2     647
3      20
Name: count, dtype: int64
```

In [37]:

```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Get scaled data from the fitted pipeline
X_scaled = pipeline.named_steps['scaler'].transform(X)

# PCA only for visualization
pca = PCA(n_components=2, random_state=42)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(10,6))

#Cluster interpretation:
cluster_names = {
    0: "Mid-Price Restaurants",
    1: "Cheap and Least-rated Restaurants",
    2: "Premium Restaurants",
    3: "Highly-rated and Customer-Satisfying Restaurants"
}

# Plot each cluster separately
for cluster in sorted(df['Restaurant Segment'].unique()):
    plt.scatter(
        X_pca[df['Restaurant Segment'] == cluster, 0],
```

```
        X_pca[df['Restaurant Segment'] == cluster, 1],
        s=50,
        label=cluster_names[cluster]
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Restaurant Segments (K-Means + PCA Visualization)")

plt.legend(title="Restaurant Segments")
plt.grid(True)
plt.show()
```



Cluster 0: Mid-Price Restaurants

These restaurants offer a good balance between affordability and quality. Their high customer ratings indicate strong customer satisfaction despite being moderately priced. Swiggy should increase their visibility through personalized recommendations, featured listings, and promotional campaigns to attract more customers and increase order volumes.

Cluster 1: Cheap and Least-rated Restaurants

These restaurants struggle with both customer satisfaction and popularity. Swiggy should work with these restaurant partners to improve food quality, customer service, and delivery efficiency before investing in promotional campaigns. Performance improvement initiatives can help increase customer trust and retention.

Cluster 2: Premium Restaurants

These restaurants cater to customers seeking premium dining experiences. Swiggy can target this segment with exclusive offers, premium memberships, and luxury dining collections. Reducing delivery time could further improve customer satisfaction and encourage repeat orders.

Cluster 3: Highly-rated and Customer-Satisfying Restaurants

These are the platform's top-performing restaurants, combining high customer satisfaction, exceptional popularity, and efficient delivery. Swiggy should prioritize these restaurants for homepage recommendations, featured campaigns, premium partnerships, and loyalty programs, as they are likely to generate the highest

featured campaigns, premium partnerships, and loyalty programs, as they are likely to generate the highest customer engagement and revenue.

In [ ]:

In [ ]:

In [ ]:

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